Project Design Phase-I

Proposed Solution Template

Date	31 January 2026
Team ID	LTVIP2026TMIDS86494
Project Name	Plugging into the Future: An Exploration of Electricity Consumption Patterns Using Tableau
Maximum Marks	2 Marks

Proposed Solution Template:

Project team shall fill the following information in proposed solution template.

S.No.	Parameter	Description
1.	Problem Statement (Problem to be solved)	Rapid urbanization, increasing dependence on electronic devices, and inefficient energy usage have led to rising electricity consumption. This results in higher costs for consumers, strain on power grids, and increased carbon emissions. There is a need for a smarter, data-driven approach to monitor, analyze, and optimize electricity consumption at household and community levels.
2.	Idea / Solution description	Develop a Smart Electricity Monitoring and Analytics System that tracks real-time energy usage using IoT-enabled smart meters. The system will provide: • Real-time consumption data via a mobile/web dashboard • AI-based usage pattern analysis • Personalized energy-saving recommendations • Peak load alerts and bill prediction • Integration with renewable energy sources (e.g., solar) The platform empowers users to make informed decisions and reduce unnecessary electricity usage.
3.	Novelty / Uniqueness	• Combines IoT + AI analytics + user-friendly visualization in one integrated platform • Predictive energy consumption forecasting • Gamified energy-saving rewards system • Community-level comparison dashboards to encourage competitive energy saving • Adaptive AI that learns user behavior and suggests automated optimizations
4.	Social Impact / Customer Satisfaction	Reduces electricity bills for households Promotes sustainable energy consumption habits

		Decreases carbon footprint and environmental impact Encourages responsible energy usage in communities Improves customer awareness and control over energy expenses
5.	Business Model (Revenue Model)	Subscription-based model (monthly/annual premium analytics plans) Sale or leasing of smart monitoring hardware Partnerships with utility companies Data insights services for energy providers (anonymized analytics) Commission from renewable energy integration services
6.	Scalability of the Solution	Cloud-based infrastructure allows nationwide and global deployment Modular IoT system adaptable for homes, offices, industries, and smart cities Easy integration with existing smart meters and grid systems Expandable to include water and gas consumption monitoring Suitable for both developed and developing regions